# Short Course on Quantile Regression

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Quantile regression is a statistical technique intended to estimate, and conduct inference about, conditional quantile functions. Just as classical linear regression methods based on minimizing sums of squared residuals enable one to estimate models for conditional mean functions, quantile regression methods offer a mechanism for estimating models for the conditional median function, and the full range of other conditional quantile functions. By supplementing the estimation of conditional mean functions with techniques for estimating an entire family of conditional quantile functions, quantile regression is capable of providing a more complete statistical analysis of the stochastic relationships among random variables.

Quantile regression has been used in a broad range of application settings. Reference growth curves for childrens' height and weight have a long history in pediatric medicine; quantile regression methods may be used to estimate upper and lower quantile reference curves as a function of age, sex, and other covariates without imposing stringent parametric assumptions on the relationships among these curves. Quantile regression methods have been widely used in economics to study determinents of wages, discrimination effects, and trends in income inequality. Several recent studies have modeled the performance of public school students on standardized exams as a function of socio-economic characteristics like their parents' income and educational attainment, and policy variables like class size, school expenditures, and teacher qualifications. It seems rather implausible that such covariate effects should all act so as to shift the entire distribution of test results by a fixed amount. It is of obvious interest to know whether policy interventions alter performance of the strongest students in the same way that weaker students are affected. Such questions are naturally investigated within the quantile regression framework.

In ecology, theory often suggests how observable covariates affect limiting sustainable population sizes, and quantile regression has been used to directly estimate models for upper quantiles of the conditional distribution rather than inferring such relationships from models based on conditional central tendency. In survival analysis, and event history analysis more generally, there is often also a desire to focus attention on particular segments of the conditional distribution, for example survival prospects of the oldest-old, without the imposition of global distributional assumptions.

## 1. Quantiles, Ranks and Optimization

We say that a student scores at the $\tau$th quantile of a standardized exam if he performs *better* than the proportion $\tau$, and *worse* than the proportion $(1-\tau)$, of the reference group of students. Thus, half of the students perform better than the median student, and half perform worse. Similarly, the quartiles divide the population into four segments with equal proportions of the population in each segment. The quintiles divide the population

into 5 equal segments; the deciles into 10 equal parts. The quantile, or percentile, refers to the general case.

More formally, any real valued random variable, $Y$, may be characterized by its distribution function,

$$F(y) = \text{Prob}(Y \leq y)$$

while for any $0 < \tau < 1$,

$$Q(\tau) = \inf\{y : F(y) \geq \tau\}$$

is called the $\tau$th quantile of $X$. The median, $Q(1/2)$, plays the central role. Like the distribution function, the quantile function provides a complete characterization of the random variable, Y.

The quantiles may be formulated as the solution to a simple optimization problem. For any $0 < \tau < 1$, define the piecewise linear "check function", $\rho_\tau(u) = u(\tau - I(u < 0))$ illustrated in Figure 1.

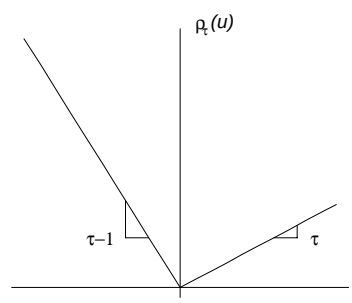


FIGURE 1. Quantile Regression $\rho$ Function

Minimizing the expectation of $\rho_\tau(Y - \xi)$ with respect to $\xi$ yields solutions, $\hat{\xi}(\tau)$, the smallest of which is $Q(\tau)$ defined above.

The sample analogue of $Q(\tau)$, based on a random sample, $\{y_1, ..., y_n\}$, of $Y$'s, is called the $\tau$th *sample* quantile, and may be found by solving,

$$\min_{\xi \in \mathbf{R}} \sum_{i=1}^{n} \rho_\tau(y_i - \xi),$$

While it is more common to define the sample quantiles in terms of the order statistics, $y_{(1)} \leq y_{(2)} \leq ... \leq y_{(n)}$, constituting a sorted rearrangement of the original sample, their formulation as a minimization problem has the advantage that it yields a natural generalization of the quantiles to the regression context.

Just as the idea of estimating the unconditional mean, viewed as the minimizer,

$$\hat{\mu} = \text{argmin}_{\mu \in \mathbb{R}} \sum (y_i - \mu)^2$$

can be extended to estimation of the linear conditional mean function $E(Y|X = x) = x'\beta$ by solving,

$$\hat{\beta} = \text{argmin}_{\beta \in \mathbb{R}^p} \sum (y_i - x_i'\beta)^2,$$

the linear conditional quantile function, $Q_Y(\tau|X = x) = x_i'\beta(\tau)$, can be estimated by solving,

$$\hat{\beta}(\tau) = \text{argmin}_{\beta \in \mathbb{R}^p} \sum \rho_\tau(y_i - x'\beta).$$

The median case, $\tau = 1/2$, which is equivalent to minimizing the sum of *absolute* values of the residuals has a long history. In the mid-18th century Boscovich proposed estimating a bivariate linear model for the ellipticity of the earth by minimizing the sum of absolute values of residuals subject to the condition that the mean residual took the value zero. Subsequent work by Laplace characterized Boscovich's estimate of the slope parameter as a weighted median and derived its asymptotic distribution. F.Y. Edgeworth seems to have been the first to suggest a general formulation of median regression involving a multivariate vector of explanatory variables, a technique he called the "plural median". The extension to quantiles other than the median was introduced in Koenker and Bassett (1978).

## 2. Two Examples

To illustrate the approach we may consider an analysis of a simple first order autoregressive model for maximum daily temperature in Melbourne, Australia. The data are taken from Hyndman, Bashtannyk, and Grunwald (1996). In Figure 2 we provide a scatter plot of 10 years of daily temperature data: today's maximum daily temperature is plotted against yesterday's maximum. Our first observation from the plot is that there is a strong tendency for data to cluster along the (dashed) 45 degree line implying that with high probability today's maximum is near yesterday's maximum. But closer examination of the plot reveals that this impression is based primarily on the left side of the plot where the central tendency of the scatter follows the 45 degree line very closely. On the right side, however, corresponding to summer conditions, the pattern is more complicated. There, it appears that *either* there is another hot day, falling again along the 45 degree line, *or* there is a dramatic cooling off. But a mild cooling off appears to be more rare. In the language of conditional densities, if today is hot, tomorrow's temperature appears to be bimodal with one mode roughly centered at today's maximum, and the other mode centered at about 20°.

Several estimated quantile regression curves have been superimposed on the scatterplot. Each curve is specified as a linear B-spline. Under winter conditions these curves are bunched around the 45 degree line, however in the summer it appears that the upper quantile curves are bunched around the 45 degree line and around 20°. In the intermediate temperatures the spacing of the quantile curves is somewhat greater indicating lower probability of this temperature range. This impression is strengthened by considering a sequence of density plots based on the quantile regression estimates. Given a family of reasonably densely spaced estimated conditional quantile functions, it is straightforward to estimate the conditional density of the response at various values of the conditioning covariate. In Figure 3 we illustrate this approach with several density estimates based

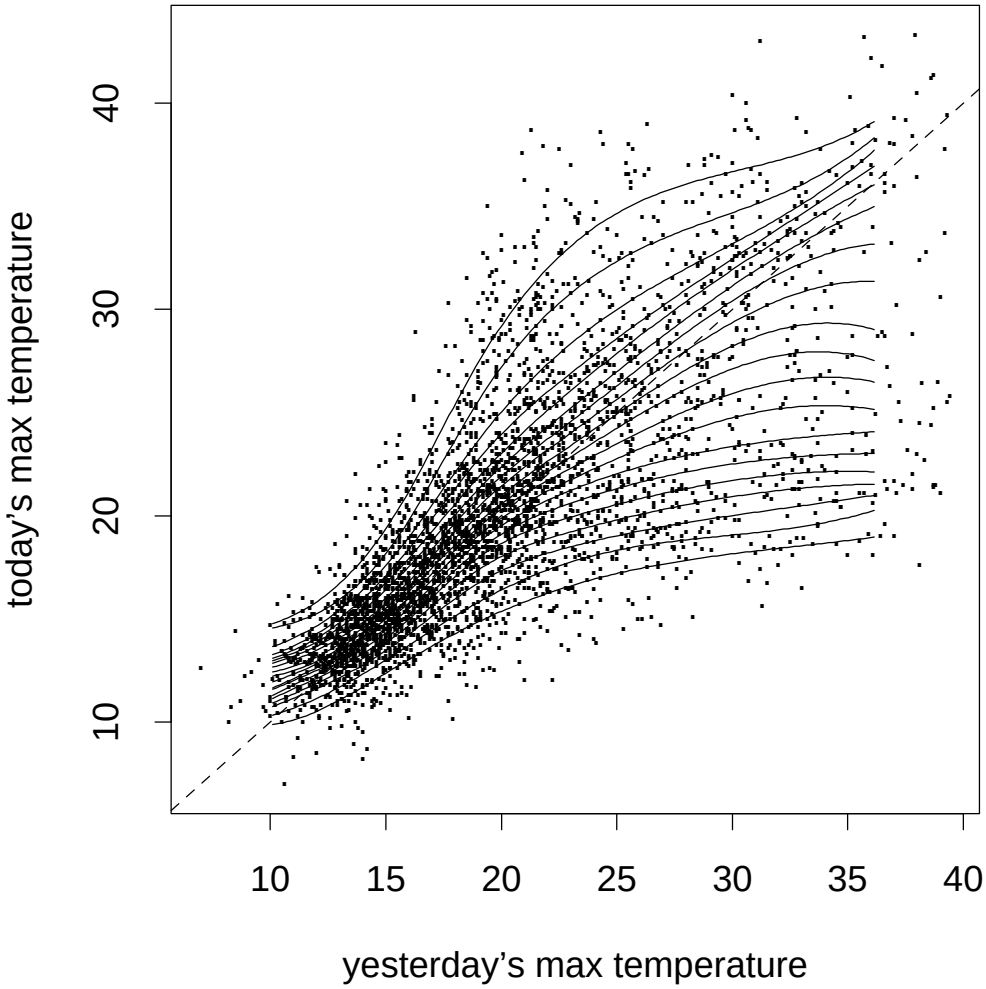


FIGURE 2. Melbourne Maximum Daily Temperature: The plot illustrates 10 years of daily maximum temperature data (in degrees centigrade) for Melbourne, Australia as an AR(1) scatterplot. The data is scattered around the (dashed) 45 degree line suggesting that today is roughly similar to yesterday. Superimposed on the scatterplot are estimated conditional quantile functions for the quantiles $\tau \in \{.05, .10, ..., .95\}$. Note that when yesterday's temperature is high the spacing between adjacent quantile curves is narrower around the 45 degree line and at about 20 degrees Centigrade than it is in the intermediate region. This suggests bimodality of the conditional density in the summer.

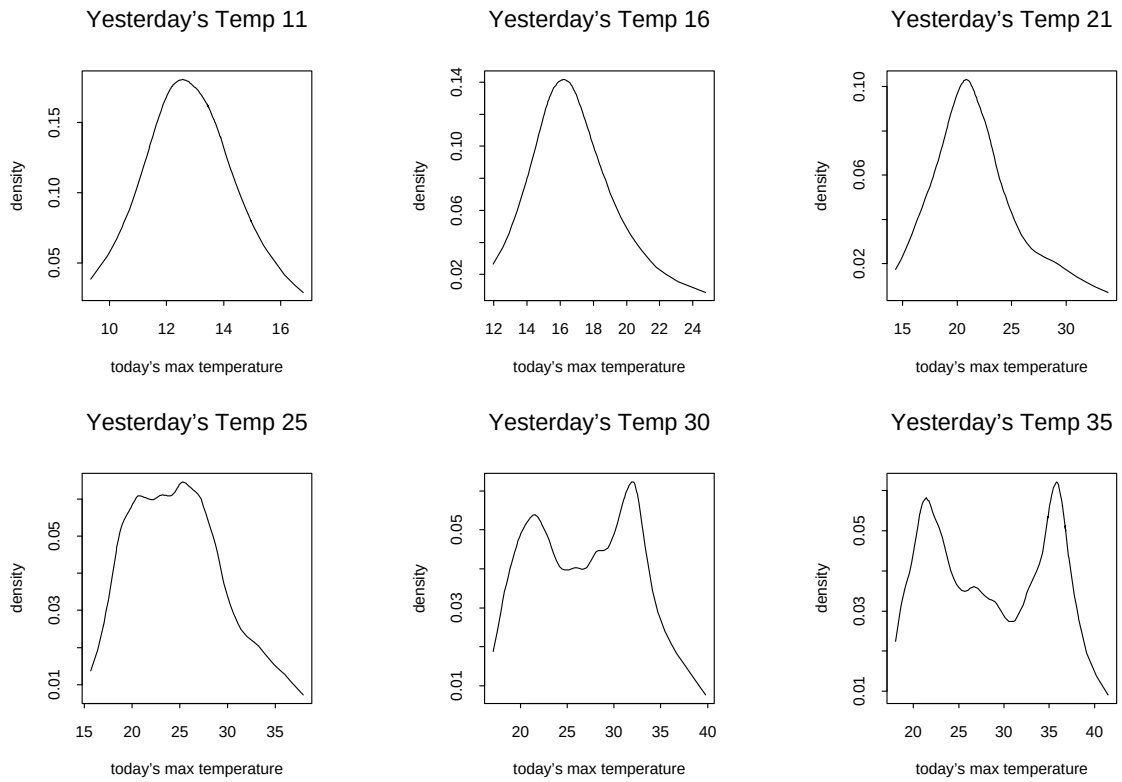


FIGURE 3. Conditional density estimates of today's maximum temperature for several values of yesterday's maximum temperature based on the Melbourne data: These density estimates are based on a kernel smoothing of the conditional quantile estimates as illustrated in the previous figure using 99 distinct quantiles. Note that temperature is bimodal when yesterday was hot.

on the Melbourne data. Conditioning on a low previous day temperature we see a nice unimodal conditional density for the following day's maximum temperature, but as the previous day's temperature increases we see a tendency for the lower tail to lengthen and eventually we see a clearly bimodal density. In this example, the classical regression assumption that the covariates affect only the location of the response distribution, but not its scale or shape, is clearly violated.

In our second example we reconsider an investigation by Abreveya(2000) of the impact of various demographic characteristics and maternal behavior on the birthweight of infants born in the U.S. Low birthweight is known to be associated with a wide range of subsequent health problems, and has even been linked to educational attainment and labor market outcomes. Consequently, there has been considerable interest in factors influencing birthweights, and public policy initiatives that might prove effective in reducing the incidence of low birthweight infants.

Although most of the analysis of birthweights has employed conventional least squares regression methods it has been recognized that the resulting estimates of various effects

on the conditional mean of birthweights were not necessarily indicative of the size and nature of these effects on the lower tail of the birthweight distribution. In an effort to focus attention more directly on the lower tail, several studies have recently explored binary response (e.g. probit) models for the occurence of low birthweights – conventionally defined to be infants weighing less than 2500 grams. Quantile regression offers a natural complement to these prior modes of analysis. A more complete picture of covariate effects can be provided by estimating a family of conditional quantile functions, as we will now illustrate.

Our analysis will be based on the June, 1997 Detailed Natality Data published by the National Center for Health Statistics. Like Abreveya (2000), we limit the sample to singleton births, with mothers recorded as either black or white, between the ages of 18 and 45, resident in the U.S. Observations with missing data for any of the variables described below were also dropped from the analysis. This process yielded a sample of 198,377 babies. Education of the mother is divided into four categories: less than high school, high school, some college, and college graduate. The omitted category is less than high school so coefficients may be interpreted relative to this category. The prenatal medical care of the mother is also divided into 4 categories: those with no prenatal visit, those whose first prenatal visit was in the first trimester of the pregnancy, those with first visit in the second trimester, and those with first visit in the last trimester. The omitted category is the group with a first visit in the first trimester; they consititute almost 85 percent of the sample. The other variables are, I hope self-explanatory.

In Figure 4 we present a concise summary of the quantile regression results for this example. Each plot depicts one coefficient in the quantile regression model. The solid line with filled dots represents the point estimates, $\{\hat{\beta}_j(\tau) \quad j = 1, ..., 16\}$, with the shaded grey area depicting a 90 percent pointwise confidence band. Superimposed on the plot is a dashed line representing the ordinary least squares estimate of the mean effect, with two dotted lines representing again a 90 percent confidence interval for this coefficient.

In the first panel of the figure the intercept of the model may be interpreted as the estimated conditional quantile function of the birthweight distribution of a girl born to an unmarried, white mother with less than a high school education, who is 27 years old and had a weight gain of 30 pounds, didn't smoke, and had her first prenatal visit in the first trimester of the pregnancy. The mother's age and weight gain are chosen to reflect the means of these variables in the sample. Note that the $\tau = .05$ quantile of this distribution is just at the margin of the conventional definition of a low birthweight baby. [This strongly suggests that it would be desirable to expand the sample and estimate models for lower quantiles.]

Boys are obviously bigger than girls, about 100 grams bigger according to the OLS estimates of the mean effect, but as is clear from the quantile regression results the disparity is much smaller in the lower quantiles of the distribution and somewhat large than 100 grams in the upper tail of the distribution. At any chosen quantile we can ask how different are the corresponding weights of boys and girls, given a specification of the other conditioning variables. The second panel answers this question.

Perhaps surprisingly, the marital status of the mother seems to be associated with a rather large positive effect on birthweight especially in the lower tail of the distribution.

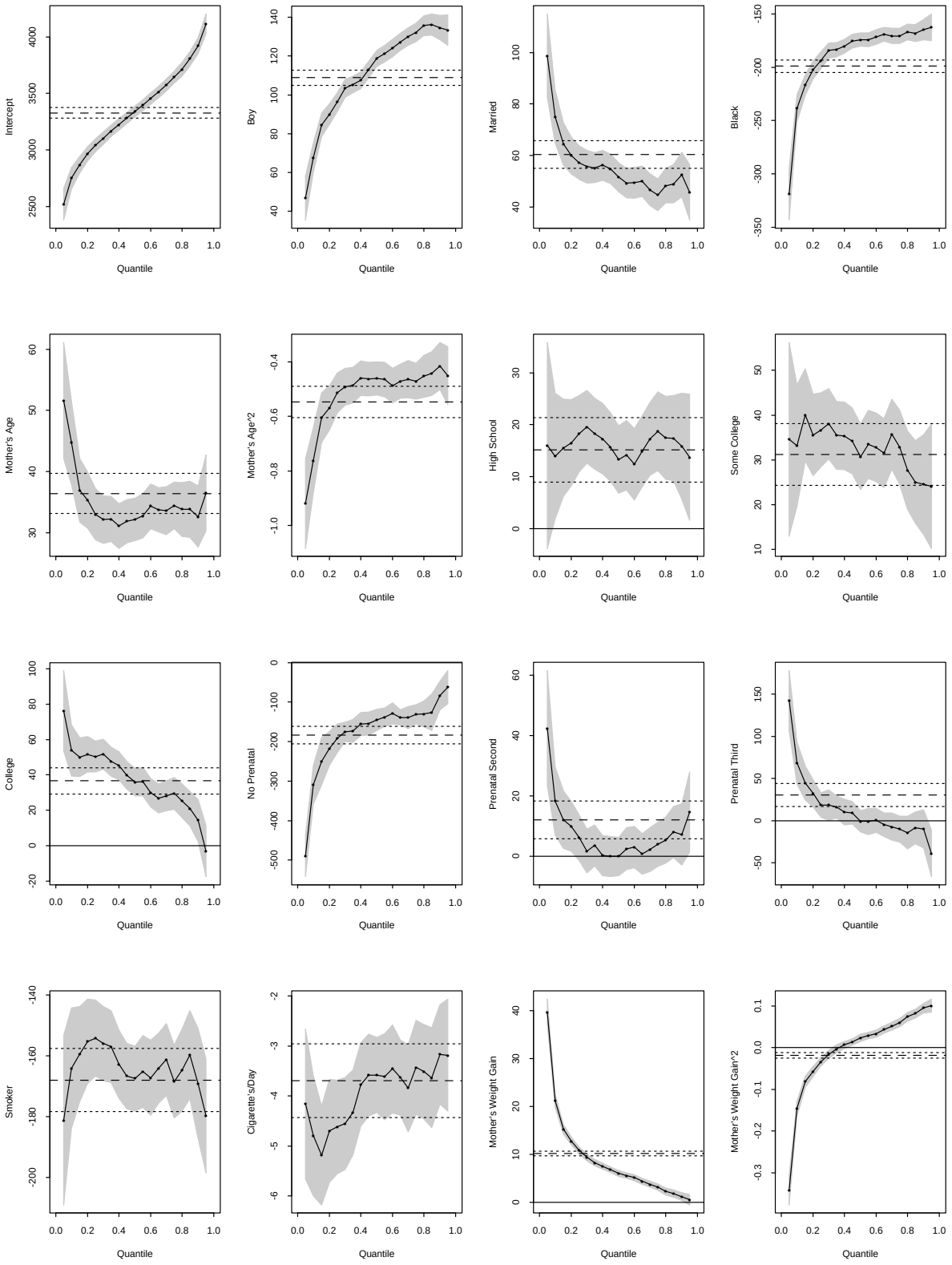


FIGURE 4. Quantile Regression for Birthweights

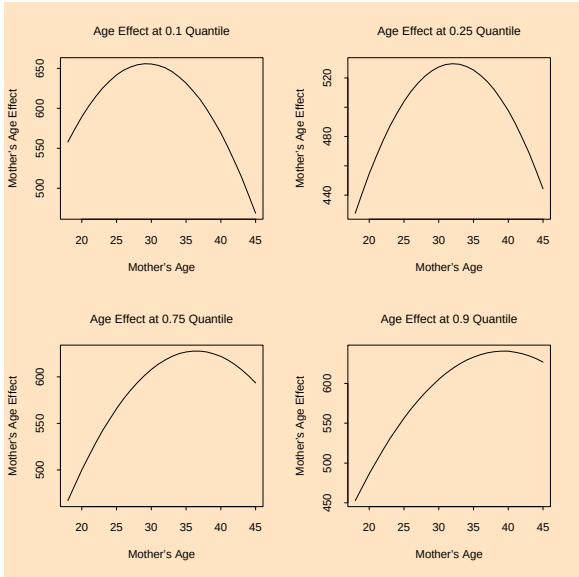


FIGURE 5. Mother's Age Effect on Birthweights: The estimated quadratic effect of mother's age on infant birthweight is illustrated at four different quantiles of the conditional birthweight distribution. In the lower tail of the conditional distribution mothers who are roughly 30 years of age have the largest children, but in the upper tail it is mothers who are 35-40 who have the largest children.

The (re)public(an) health implications of this finding should, of course, be viewed with caution, however.

The disparity between birthweights of infants born to black and white mothers is very large particularly at the left tail of the distribution. The difference in birth weight between a baby born to a black mother and a white mother at the 5th percentile of the conditional distribution is roughly one third of a kilogram.

Mother's age enters the model as a quadratic. At the lower quantiles the mother's age tends to be more concave, increasing birthweight from age 18 to about age 30, but tending to decrease birthweight when the mother's age is beyond 30. At higher quantiles there is also this optimal age, but it becomes gradually older. At the third quantile it is about 36, and at $\tau = .9$ it is almost 40. This is illustrated in Figure 5.

Education beyond high school is associated with a modest increase in birthweights. High school graduation has a quite uniform effect over the whole range of the distribution of about 15 grams. This is a rare example of an effect that really does appear to exert a pure location shift effect on the conditional distribution. Some college education has a somewhat more positive effect in the lower tail than in the upper tail, varying from about 35 grams in the lower tail to 25 grams in the upper tail. A college degree has an even more substantial positive effect, but again much larger in the lower tail and declining to a negligble effect in the upper tail.

The effect of prenatal care is of obvious public health policy interest. Since individuals self-select into prenatal care results must be interpreted with considerable caution. Those

receiving no prenatal care are likely to be at risk in other dimensions as well. Nevertheless, the effects are sufficiently large to warrent considerable further investigation. Babies born to mothers who received no prenatal care were on average about 150 grams lighter than those who had a prenatal visit in the first trimester. In the lower tail of the distribution this effect is considerably larger – at the 5th percentile it is nearly half a kilogram! In contrast, mothers who delayed prenatal visits until the second or third trimester have substantially *higher* birthweights in the lower tail than mothers who had a visit in the first trimester. This might be interpreted as the self-selection effect of mothers confident about favorable outcomes. In the upper 3/4 of the distribution there seems to be no significant effect.

Smoking has a clearly deleterious effect. The indicator of whether the mother smoked during the pregnancy is associated with a decrease of about 175 grams in birthweight. In addition, there is an effect of about 4 to 5 grams per cigarette per day. Thus a mother smoking a pack per day appears to induce a weight reduction of about 250 to 300 grams in their babies.

Lest this smoking effect be thought to be attributable to some associated reduction in the mothers weight gain, we should hasten to point out that the weight gain effect is explicitly accounted for with a quadratic specification. Not suprisingly, the mother's weight gain has a very strong influence on birthweight, and this is reflected in the very narrow confidence band for both linear and quadratic coefficients. In Figure 6 we illustrate this marginal effect of weight gain by evaluating over the entire range of quantiles for four different levels of weight gain. At low weight gains by the mother the marginal effect of another pound gained is about 30 grams at the lowest quantiles and declines to only about 5 grams at the upper quantiles. This pattern of declining marginal effects is maintained for large weight gains until we begin to consider extremely large weight gains at which point the effect is reversed. For example, another pound gained by the mother who has already gained 50 pounds has only a 7 gram effect in the lower tail of the birthweight distribution and this increases to about 10 grams at the upper quantiles. The quadratic specification of the effect of mother's weight gain offers a striking example of how misleading the OLS estimates can be. Note that the OLS estimates strongly suggest that the effect is linear with an essentially negligible quadratic effect. However, the quantile regression estimates give a very different picture, one in which the quadratic effect of the weight gain is very significant except where it cross the zero axis at about $\tau = .33$.

Although much more could be drawn out of the foregoing analysis, it may suffice to conclude here with the comment that the quantile regression results offer a much richer, more focused view of both of the applications than can be achieved by exclusively looking at conditional mean models.

## 3. Interpretation of Quantile Regression

Least squares estimation of mean regression models asks the question, "How does the conditional mean of $Y$ depend on the covariates $X$?" Quantile regression asks this question at each quantile of the conditional distribution enabling one to obtain a more complete description of how the conditional distribution of $Y$ given $X = x$ depends on $x$. Rather

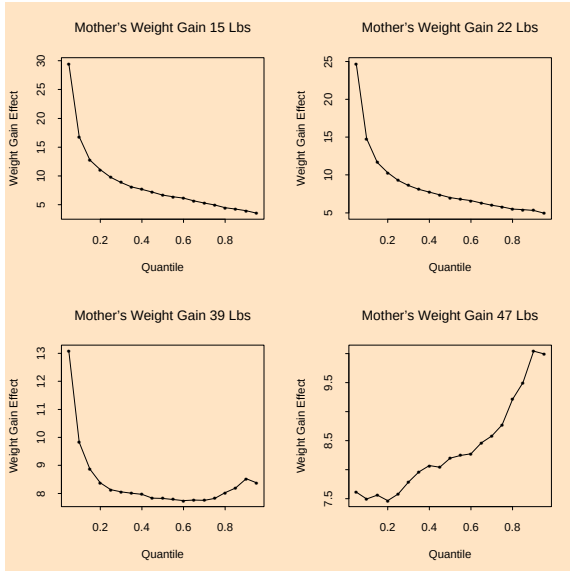


FIGURE 6. Mother's Weight Gain Marginal Effect: The marginal effect of the mother's weight gain, again parameterized as a quadratic effect, tends to decrease over the entire range of the conditional distibution of birthweights. Thus incremental weight gain is most influential in increasing the weight of low birthweight infants. But for mothers with unusually large weight gains, this pattern is reversed and effect is largest in the upper tail of the conditional birthweight distribution.

than assuming that covariates shift only the location or scale of the conditional distribution, quantile regression methods enable one to explore potential effects on the shape of the distribution as well. Thus, for example, the effect of a job-training program on the length of participants' current unemployment spell might be to lengthen the shortest spells while dramatically reducing the probability of very long spells. The mean treatment effect in such circumstances might be small, but the treatment effect on the shape of the distribution of unemployment durations could, nevertheless, be quite significant.

3.1. **Quantile Treatment Effects.** The simplest formulation of quantile regression is the two-sample treatment-control model. In place of the classical Fisherian experimental design model in which the treatment induces a simple location shift of the response distribution, Lehmann (1974) proposed the following general model of treatment response:

> "Suppose the treatment adds the amount $\Delta(x)$ when the response of the untreated subject would be $x$. Then the distribution $G$ of the treatment responses is that of the random variable $X + \Delta(X)$ where $X$ is distributed according to $F$."

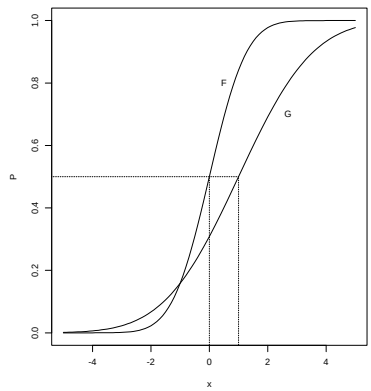


FIGURE 7. Lehmann Quantile Treatment Effect: the horizontal distannce between the treatment and control (marginal) distribution functions.

Special cases obviously include the location shift model, $\Delta(X) = \Delta_0$, and the scale shift model, $\Delta(X) = \Delta_0 X$, but the general case is natural within the quantile regression paradigm.

Doksum (1974) shows that if $\Delta(x)$ is defined as the "horizontal distance" between $F$ and $G$ at $x$, so

$$F(x) = G(x + \Delta(x))$$

then $\Delta(x)$ is uniquely defined and can be expressed as

$$\Delta(x) = G^{-1}(F(x)) - x.$$

Changing variables so $\tau = F(x)$ one may define the *quantile treatment effect*,

$$\delta(\tau) = \Delta(F^{-1}(\tau)) = G^{-1}(\tau) - F^{-1}(\tau).$$

In the two sample setting this quantity is naturally estimable by

$$\hat{\delta}(\tau) = \hat{G}_n^{-1}(\tau) - \hat{F}_m^{-1}(\tau)$$

where $G_n$ and $F_m$ denote the empirical distribution functions of the treatment and control observations, based on $n$ and $m$ observations respectively.

Formulating the quantile regression model for the binary treatment problem as,

$$Q_{Y_i}(\tau|D_i) = \alpha(\tau) + \delta(\tau)D_i$$

where $D_i$ denotes the treatment indicator, with $D_i = 1$ indicating treatment, $D_i = 0$, control, then the quantile treatment effect can be estimated by solving,

$$(\hat{\alpha}(\tau), \hat{\delta}(\tau))' = \text{argmin} \sum_{i=1}^{n} \rho_\tau(y_i - \alpha - \delta D_i).$$

The solution $(\hat{\alpha}(\tau),\ \hat{\delta}(\tau))'$ yields $\hat{\alpha}(\tau) = \hat{F}_n^{-1}(\tau)$, corresponding to the control sample, and

$$\hat{\delta}(\tau) = \hat{G}_n^{-1}(\tau) - \hat{F}_n^{-1}(\tau).$$

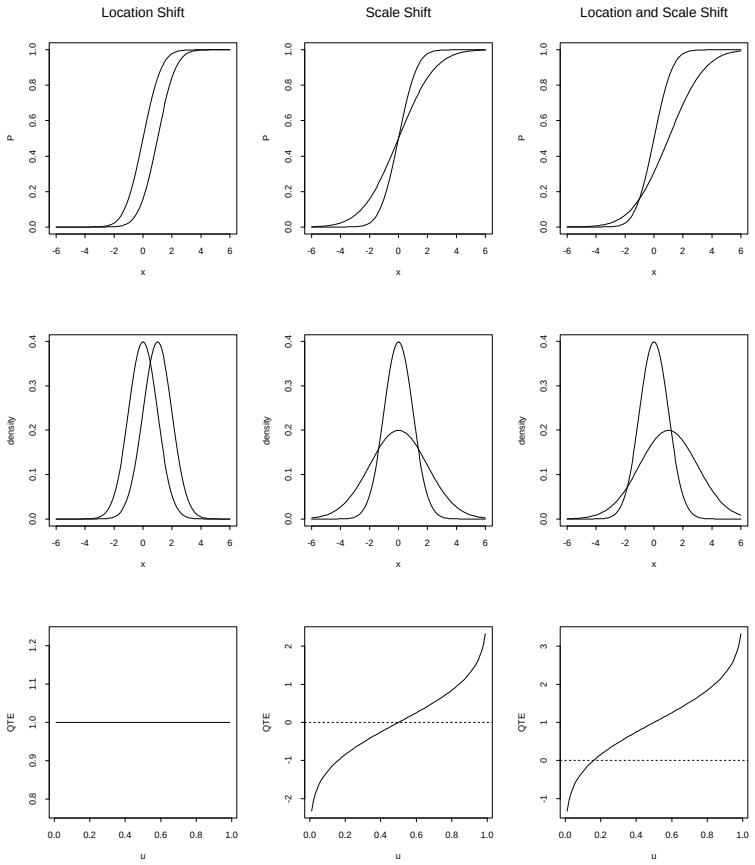


FIGURE 8. Lehmann Quantile Treatment Effect for 3 Examples: Location Shift, Scale Shift, and Location-Scale Shift.

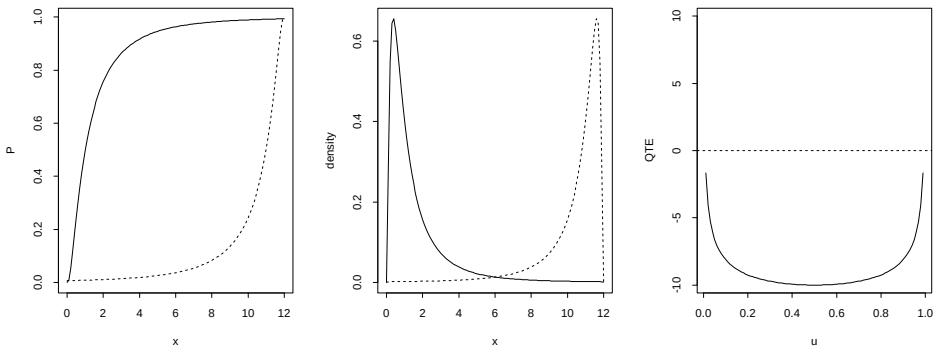


FIGURE 9. Lehmann Quantile Treatment Effect For An Asymmetric Example: the treatment reverses the skewness of the distribution function.

Doksum suggests that one may interpret control subjects in terms of a latent characteristic: for example in survival analysis applications, a control subject may be called frail if he is prone to die at an early age, and robust if he is prone to die at an advanced age. This latent characteristic is thus implicitly indexed by $\tau$, the quantile of the survival distribution at which the subject would appear if untreated, i.e., $(Y_i|D_i = 0) = \alpha(\tau)$. And the treatment, under the Lehmann model, is assumed to alter the subjects control response, $\alpha(\tau)$, making it $\alpha(\tau) + \delta(\tau)$ under the treatment. If the latent characteristic, say, the propensity for longevity, were observable *ex ante*, then one could view the treatment effect $\delta(\tau)$ as an explicit interaction with this observable variable. In the absence of such an observable variable however, the quantile treatment effect may be regarded as a natural measure of the treatment response.

It may be noted that the quantile treatment effect is intimately tied to the two-sample QQ-plot which has a long history as a graphical diagnostic device. The function $\hat{\Delta}(x) = G_n^{-1}(F_m(x)) - x$ is exactly what is plotted in the traditional two sample QQ-plot. If $F$ and $G$ are identical then the function $G_n^{-1}(F_m(x))$ will lie along the 45 degree line; if they differ only by a location scale shift, then $G_n^{-1}(F_m(x))$ will lie along another line with intercept and slope determined by the location and scale shift, respectively. Quantile regression may be seen as a means of extending the two-sample QQ plot and related methods to general regression settings with continuous covariates.

When the treatment variable takes more than two values, the Lehmann-Doksum quantile treatment effect requires only minor reinterpretation. If the treatment variable is continuous as, for example, in dose-response studies, then it is natural to consider the assumption that its effect is linear, and write,

$$Q_{Y_i}(\tau|x_i) = \alpha(\tau) + \beta(\tau)x_i.$$

We assume thereby that the treatment effect, $\beta(\tau)$, of changing $x$ from $x_0$ to $x_0 + 1$ is the same as the treatment effect of an alteration of $x$ from $x_1$ to $x_1 + 1$. Note that this notion of the quantile treatment effect measures, for each $\tau$, the change in the response required to stay on the $\tau$th conditional quantile function.

3.2. **Transformation Equivariance of Quantile Regression.** An important property of the quantile regression model is that, for any monotone function, $h(\cdot)$,

$$Q_{h(T)}(\tau|x) = h(Q_T(\tau|x)).$$

This follows immediately from observing that

$$\text{Prob}(T < t|x) = \text{Prob}(h(T) < h(t)|x).$$

This equivariance to monotone transformations of the conditional quantile function is a crucial feature, allowing one to decouple the potentially conflicting objectives of transformations of the response variable. This equivariance property is in direct contrast to the inherent conflicts in estimating transformation models for conditional mean relationships. Since, in general, $E(h(T)|x) \neq h(E(T|x))$ the transformation alters in a fundamental way what is being estimated in ordinary least squares regression.

A particularly important application of this equivariance result, and one that has proven extremely influential in the econometric application of quantile regression, involves censoring of the observed response variable. The simplest model of censoring may be formulated

as follows. Let $y_i^*$ denote a latent (unobservable) response assumed to be generated from the linear model

$$y_i^* = x_i'\beta + u_i \quad i = 1, \ldots, n$$

with $\{u_i\}$ iid from distribution function F. Due to censoring, the $y_i^*$'s are not observed directly, but instead one observe

$$y_i = \max\{0, y_i^*\}.$$

Powell (1986) noted that the equivariance of the quantiles to monotone transformations implied that in this model the conditional quantile functions of the response depended only on the censoring point, but were independent of $F$. Formally, the $\tau$th conditional quantile function of the observed response, $y_i$, in this model may be expressed as

$$Q_i(\tau|x_i) = \max\{0, x_i'\beta + F_u^{-1}(\tau)\}$$

The parameters of the conditional quantile functions may now be estimated by solving

$$\min_b \sum_{i=1}^n \rho_\tau(y_i - \max\{0, x_i'b\})$$

where it is assumed that the design vectors $x_i$, contain an intercept to absorb the additive effect of $F_u^{-1}(\tau)$. This model is computationally somewhat more demanding than conventional linear quantile regression because it is non-linear in parameters.

3.3. **Robustness.** Robustness to distributional assumptions is an important consideration throughout statistics, so it is important to emphasize that quantile regression inherits certain robustness properties of the ordinary sample quantiles. The estimates and the associated inference apparatus have an inherent distribution-free character since quantile estimation is influenced only by the local behavior of the conditional distribution of the response near the specified quantile. Given a solution $\hat{\beta}(\tau)$, based on observations, $\{y, X\}$, as long as one doesn't alter the sign of the residuals, *any* of the $y$ observations may be arbitrary altered without altering the initial solution. Only the signs of the residuals matter in determining the quantile regression estimates, and thus outlying responses influence the fit in so far as they are either above or below the fitted hyperplane, but how far above or below is irrelevant.

While quantile regression estimates are inherently robust to contamination of the response observations, they can be quite sensitive to contamination of the design observations, $\{x_i\}$. Several proposals have been made to ameliorate this effect.

## 4. Computational Aspects of Quantile Regression

Although it was recognized by a number of early authors, including Gauss, that solutions to the median regression problem were characterized by an exact fit through $p$ sample observations when $p$ linear parameters are estimated, no effective algorithm arose until the development of linear programming in the 1940's. It was then quickly recognized that the median regression problem could be formulated as a linear program, and the simplex method employed to solve it. The algorithm of Barrodale and Roberts (1973) provided the first efficient implementation specifically designed for median regression and is still widely used in statistical software. It can be concisely described as follows. At

each step, we have a trial set of $p$ "basic observations" whose exact fit may constitute a solution. We compute the directional derivative of the objective function in each of the $2p$ directions that correspond to removing one of the current basic observations, and taking either a positive or negative step. If none of these directional derivatives are negative the solution has been found, otherwise one chooses the most negative, the direction of steapest descent, and goes in that direction until the objective function ceases to decrease. This one dimensional search can be formulated as a problem of finding the solution to a scalar weighted quantile problem. Having chosen the step length, we have in effect determined a new observation to enter the basic set, a simplex pivot occurs to update the current solution, and the iteration continues.

This modified simplex strategy is highly effective on problems with a modest number of observations, achieving speeds comparable to the corresponding least squares solutions. But for larger problems with, say $n > 100,000$ observations, the simplex approach eventually becomes considerably slower than least squares. For large problems recent development of interior point methods for linear programming problems are highly effective. Portnoy and Koenker (1997) describe an approach that combines some statistical preprocessing with interior point methods and achieves comparable performance to least squares solutions even in very large problems.

An important feature of the linear programming formulation of quantile regression is that the entire range of solutions for $\tau \in (0,1)$ can be efficiently computed by parametric programming. At any solution $\hat{\beta}(\tau_0)$ there is an interval of $\tau$'s over which this solution remains optimal, it is straightforward to compute the endpoints of this interval, and thus one can solve iteratively for the entire sample path $\hat{\beta}(\tau)$ by making one simplex pivot at each of the endpoints of these intervals.

## 5. Statistical Inference for Quantile Regression

The asymptotic behavior of the quantile regression process $\{\hat{\beta}(\tau) : \tau \in (0,1)\}$ closely parallels the theory of ordinary sample quantiles in the one sample problem. Koenker and Bassett (1978) show that in the classical linear model,

$$y_i = x_i\beta + u_i$$

with $u_i$ iid from $dfF$, with density $f(u) > 0$ on its support $\{u|0 < F(u) < 1\}$, the joint distribution of $\sqrt{n}(\hat{\beta}_n(\tau_i) - \beta(\tau_i))_{i-1}^m$ is asymptotically normal with mean 0 and covariance matrix $\Omega \otimes D^{-1}$. Here $\beta(\tau) = \beta + F_u^{-1}(\tau)e_1, e_1 = (1,0,\ldots,0)', x_{1i} \equiv 1, n^{-1}\sum x_i x_i' \to D$, a positive definite matrix, and

$$\Omega = (\omega_{ij} = (\min\{\tau_i, \tau_j\} - \tau_i\tau_j)/(f(F^{-1}(\tau_i))f(F^{-1}(\tau_j))).$$

When the response is conditionally independent over $i$, but not identically distributed, the asymptotic covariance matrix of $\xi(\tau) = \sqrt{n}(\hat{\beta}(\tau) - \beta(\tau))$ is somewhat more complicated. Let

$$\xi_i(\tau) = x_i\beta(\tau)$$

denote the conditional quantile function of $y$ given $x_i$, and $f_i(\cdot)$ the corresponding conditional density, and define,

$$J_n(\tau_1, \tau_2) = (\min\{\tau_1, \tau_2\} - \tau_1\tau_2)n^{-1}\sum_{i=1}^{n} x_i x_i',$$

and

$$H_n(\tau) = n^{-1}\sum x_i x_i' f_i(\xi_i(\tau)).$$

Under mild regularity conditions on the $\{f_i\}$'s and $\{x_i\}$'s, we have joint asymptotic normality for vectors $(\xi(\tau_i), \ldots, \xi(\tau_m))$ with mean zero and covariance matrix

$$V_n = (H_n(\tau_i)^{-1} J_n(\tau_i, \tau_j) H_n(\tau_j)^{-1})_{i=1}^{m}.$$

An important link to the classical theory of rank tests was made by Gutenbrunner and Jurečková (1992), who showed that the rankscore functions of Hájek and Šidák (1967) could be viewed as a special case of a more general formulation for the linear quantile regression model. The formal dual of the quantile regression linear programming problem may be expressed as,

$$\max\{y'a | X'a = (1-t)X'1,\ a \in [0,1]^n\}.$$

The dual solution $\hat{a}(\tau)$ reduces to the Hájek and Šidák rankscores process when the design matrix, $X$, takes the simple form of an $n$ vector of ones. The regression rankscore process $\hat{a}(\tau)$ behaves asymptotically much like the classical univariate rankscore process, and thus offers a way to extend many rank based inference procedures to the more general regression context.

## 6. Extensions and Future Developments

There is considerable scope for further development of quantile regression methods. Applications to survival analysis and time-series modeling seem particularly attractive, where censoring and recursive estimation pose, respectively, interesting challenges. For the classical latent variable form of the binary response model where,

$$y_i = I(x_i'\beta + u_i \geq 0)$$

and the median of $u_i$ conditional on $x_i$ is assumed to be zero for all $i = 1, ..., n$, Manski (1975) proposed an estimator solving,

$$\max_{||b||=1} \sum (y_i - 1/2) I(x_i' b \geq 0).$$

This "maximum score" estimator can be viewed as a median version of the general linear quantile regression estimator for binary response,

$$\min_{||b||=1} \sum \rho_\tau(y_i - I(x_i' b \geq 0)).$$

In this formulation it is possible to estimate a family of quantile regression models and explore, semi-parametrically, a full range of linear conditional quantile functions for the latent variable form of the binary response model. This has recently been explored in considerable depth in Kordas (2001).

Koenker and Machado (1999) and Koenker and Xiao (2001) introduce inference methods closely related to classical goodness of fit statistics based on the full quantile regression process. There have been several proposals dealing with generalizations of quantile regression to nonparametric response functions involving both local polynomial methods and splines. Work by Chaudhuri (1991) and Welsh (1996) considers locally polynomial methods, while Koenker, Ng and Portnoy (1994) and Koenker and Mizera (2002) explore penalty methods. Extension of quantile regression methods to multivariate response models is a particularly important challenge.

## 7. Conclusion

Classical least squares regression may be viewed as a natural way of extending the idea of estimating an unconditional mean parameter to the problem of estimating conditional mean *functions*; the crucial step is the formulation of an optimization problem that encompasses both problems. Likewise, quantile regression offers an extension of univariate quantile estimation to estimation of conditional quantile functions via an optimization of a piecewise linear objective function in the residuals. Median regression minimizes the sum of absolute residuals, an idea introduced by Boscovich in the 18th century.

The asymptotic theory of quantile regression closely parallels the theory of the univariate sample quantiles; computation of quantile regression estimators may be formulated as a linear programming problem and efficiently solved by simplex or barrier methods. A close link to rank based inference has been forged from the theory of the dual regression quantile process, or regression rankscore process.

Recent non-technical introductions to quantile regression are provided by Buchinsky (1998) and Koenker and Hallock (2001). Related papers are available on my website `http://www.econ.uiuc.edu/~roger`. Most of the major statistical computing languages now include some capabilities for quantile regression estimation and inference. Quantile regression packages are available for the related languages R and Splus from the R archives at `http://lib.stat.cmu.edu/R/CRAN` and Statlib at `http://lib.stat.cmu.edu/S`, respectively. Stata also provides some quantile regression estimation and inference functions.

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